

Prescribing finite-window reaction-time modes with a static fixed-budget Doi reactivity field

Xiaoxiao Zhouyi*

*School of Engineering Mathematics and Technology,
University of Bristol, Bristol BS8 1TW, United Kingdom*

(Dated: September 3, 2026)

Diffusing molecules that react on contact do so at random times, and the distribution of these reaction times can have several peaks. We ask whether a fixed amount of static catalyst, installed once in space and never switched, can be arranged so that the reaction-time density of a diffusing pair has exactly a prescribed number of peaks on a chosen time window. For a two-particle Doi process—two particles that react at a position-dependent rate whenever they are in contact—we prove that for every fixed finite dimension $d \geq 2$ and mode count $m \geq 1$, and every sufficiently small budget $B > 0$, there is a nonnegative, time-independent reactivity field of fixed integrated strength $\int \kappa(c) dc = B$ whose reaction-time density has exactly m nondegenerate maxima and $m - 1$ nondegenerate minima on a declared compact positive-time window. The construction places m narrow catalyst slabs along the monotone mean trajectory of the pair’s centre as it relaxes in a common harmonic trap, so that each slab acts as a separated Gaussian exposure clock; a global bound on the number of derivative zeros and a whole-window sign certificate fix the complete stationary signature uniformly over a compact family of positive slab weights, and C^2 convergence of the budget-rescaled density on positive times transfers it to the killed process for all sufficiently small B . Off-lattice simulations realize the design at moderate noise and order-one budgets and event mass, with up to five prescribed peaks in $d = 2$ and a two-peak check in $d = 3$; they map the noise–budget boundary beyond which modes are lost and show that loss occurs by peak merger or by depletion, the two failure directions indicated by the proof margins, and a mean-field hazard–survival law built from the free exposure clock reproduces that boundary to within about 5% with no adjustable parameter. The theorem is an existence statement with a sequential quantifier order: it supplies neither a budget uniform in d and m nor any statement about topology outside the chosen window.

I. INTRODUCTION

Many diffusion-controlled processes [1–4] are read out through the time at which a reaction first occurs: the escape of a molecule through a small receptor patch on a cell membrane [5, 6], target search in the crowded cell [7–10], and catalysis on heterogeneous surfaces [11]. In all of these the full reaction-time distribution, not only its mean, carries the information [12–14], and a multi-peaked distribution encodes several distinct pathways or timings. The inverse question is a design question: given a fixed total amount of reactivity—a catalyst load or surface coverage installed once and never modulated in time—can the reaction-time density be made to have a prescribed number of peaks? This paper answers that question constructively for a pair of diffusing particles that react at a position-dependent rate whenever they are in contact, the Doi model [15–17].

Reaction-time multimodality is established rather than new. Reaction-time densities in confinement are typically broad and carry several time scales [12, 18]; strategically arranged static traps can generate multimodal capture-time densities [19]; and distinct transport channels, heterogeneous reaction control, and encounter geometries can create separated time scales [14, 20–23]. The question here is therefore not whether several peaks can occur,

but whether their complete finite-window stationary-point topology can be constructed under a specific static resource constraint.

Several adjacent inverse-design paradigms clarify that constraint. Classical inverse first-passage theory uses a time-dependent absorbing boundary to realize a prescribed survival law [24, 25]. Soft-killing variants translate a killing profile by a time-dependent barrier and establish existence or uniqueness for prescribed lifetime distributions [26–28]. Finite-state phase-type and hyper-Erlang constructions generate flexible positive waiting-time laws by introducing hidden Markov phases [29, 30]. Stochastic gating modifies detection times through switching reaction states [31–33], and kinetic proofreading uses sequential, energy-consuming reaction steps to impose temporal delays [34–36]. These results provide broader distributional control in different design spaces; they do not impose the static spatial constraint considered here. Linear-response theory predicts how first-passage statistics change under small perturbations [37]; the question here is instead constructive: which static field realizes a prescribed peak count.

Static spatial reactivity under a fixed total resource is itself also established. Heterogeneous bulk killing has been studied with fixed total reactive strength, partially reactive network elements have been rearranged at fixed total absorbance, and position-dependent Robin reactivity has been optimized under a fixed integral to maximize steady flux [38–40]. The objective here is different: the exhaustive transient stationary-point topology of a two-

* xiaoxiao.zhouyi@bristol.ac.uk

particle Doi reaction-time density under such a constraint, together with its whole-window positive-budget transfer.

Our admissible control is a lab-frame, time-independent, nonnegative centre-coordinate field $\kappa(c)$, multiplied by a fixed encounter-contact profile and constrained by

$$\int \kappa(c) dc = B. \quad (1)$$

Every admissible allocation uses the same installed amount B . We neither translate nor activate the field in time, introduce signed reactivity, nor add an internal gating or proofreading state. Under these restrictions we ask whether a prescribed number of reaction-time modes can nevertheless be made exact on a specified time window. Table I states the resulting construction as a five-step recipe.

Main result.—In words: for any fixed dimension and any target number m of peaks, one can place m narrow catalyst slabs of fixed total strength B along the path of the pair’s centre so that, once the noise-and-width scale ε is small enough and then B is small enough, the reaction-time density on the design window has exactly m peaks and $m - 1$ troughs; this holds for every allocation of the total strength that gives each slab at least a fixed fraction. The free-exposure clock used below is the budget-rescaled density f_B/B in the limit $B \rightarrow 0$, i.e. the rate at which the unkilld pair is exposed to the catalyst. Formally: fix finite $d \geq 2$, finite $m \geq 1$, a compact interval $I = [\tau, T] \subset (0, \infty)$, and a weight floor $0 < w_* \leq 1/m$. We construct transport and initial data, contact geometry, and m ordered normalized catalyst slabs; these data may depend on (d, m) . There is $\varepsilon_0 > 0$, depending on these data and on w_* , such that, for every $0 < \varepsilon < \varepsilon_0$, the free-exposure clock has exactly m nondegenerate interior maxima and $m - 1$ nondegenerate interior minima on I and no stationary endpoint, uniformly for all slab weights satisfying $w_j \geq w_*$ and $\sum_j w_j = 1$. For each fixed such ε , there is a positive uniform topological threshold $B_{\text{top}}^{\text{unif}}(\varepsilon)$ such that the positive-budget Doi reaction-time density has the identical stationary signature whenever $0 < B < B_{\text{top}}^{\text{unif}}(\varepsilon)$. For every allocation the field remains static and satisfies Eq. (1).

The result shows that a static, non-switchable resource of fixed integrated strength suffices to prescribe the complete finite-window peak–valley structure of a reaction-time density, without the time-dependent barriers, gating states, or hidden kinetic phases used by the inverse-design paradigms above. The upper bound of m modes for an m -component univariate Gaussian mixture is classical [41, 42]; the proof burden created by the encounter process—constructing all $2m - 1$ stationary points with uniform margins together with a global derivative-zero bound, excluding further zeros over the whole window after multiplication by the contact probability, and preserving the root tubes (disjoint neighbourhoods, one around each stationary point), curvature signs, endpoint slopes, and complement margin (a nonzero derivative bound outside the tubes) under positive-budget Doi killing—is taken

up in Sec. III, with the last transfer supplied by convergence of the budget-rescaled density together with its time and allocation derivatives, uniformly on the window (Sec. III A).

The theorem is an existence statement with a sequential quantifier order (Sec. III). The off-lattice campaigns of Sec. IV therefore test its finite-parameter content directly: realizability at order-one budgets, the operational noise–budget retention boundary, and the predicted peak-merger and depletion failures, all judged by a classifier fixed in advance. Doi killing, reaction mass balance, and spatially heterogeneous reactivity are established ingredients [11, 15–17, 43–47].

a. Physical picture. Two particles subject to the same longitudinal harmonic trap start displaced from the trap centre and in contact, and their midpoint relaxes past m static catalyst slabs of fixed total strength B . Because the slabs are static, the only clock is the relaxation itself: the field converts the position of the pair into a reaction time, and the modes are the image of the slab pattern under this map. In the theorem regime the deterministic relative trajectory stays inside the contact radius on the whole window and the contact probability tends to one as $\varepsilon \rightarrow 0$ (Supplemental Material, Lemma S3.1), so the encounter gate is a slow positive factor and, at leading order, the pair acts as a single reactive complex sampling the striped hazard $BG(t)$; the exact Doi law differs from this time-inhomogeneous Poisson clock through the fluctuation of each walker’s exposure and its correlation with position, which Theorem 1 controls in the small- B limit and which the simulations of Sec. IV bound at order-one B . A pair started at the trap centre would sample a time-independent slab profile: the static constraint is on the reactivity, and the transient supplies the time dependence.

II. CONSERVED-REACTIVITY ENCOUNTER MODEL

Let $q(x, t)$ be the unreacted configuration density of two mobile particles (an encounter problem [23, 48]), with conservative generator $\mathcal{L}$. Write r for the relative coordinate and c for an encounter-centre coordinate. For a fixed contact profile $\chi_a(r)$ and m normalized nonnegative centre-space profiles $\phi_j(c)$, $j = 1, \dots, m$, define

$$\begin{aligned} \kappa_{\mathbf{w}}(c) &= B \sum_{j=1}^m w_j \phi_j(c), & w_j &\geq 0, \\ \sum_j w_j &= 1, & \int \phi_j(c) dc &= 1, \\ K_{B, \mathbf{w}}(x) &= \chi_a(r) \kappa_{\mathbf{w}}(c). \end{aligned} \quad (2)$$

Every admissible allocation has the same installed centre-space resource,

$$\int \kappa_{\mathbf{w}}(c) dc = B. \quad (3)$$

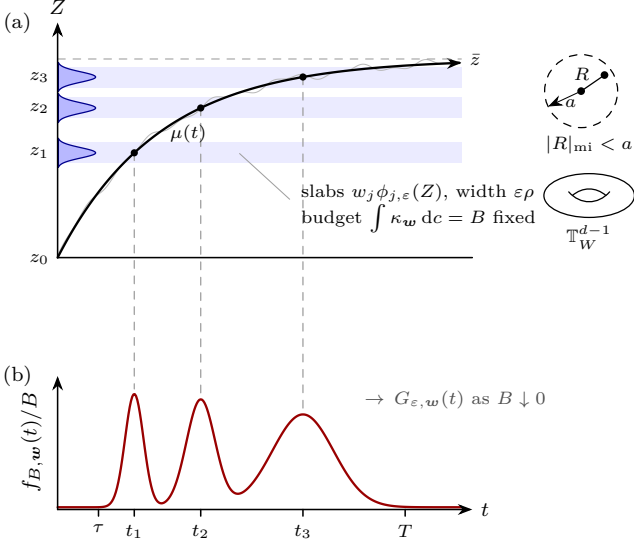


FIG. 1. Mechanism of prescribed finite modality ($m = 3$ shown). (a) The midpoint mean $\mu(t) = \bar{z} + (z_0 - \bar{z})e^{-\gamma t}$ drifts monotonically toward $\bar{z}$ (thick curve; a sample path of Z_t in grey) past three narrow normalized Gaussian catalyst slabs $w_j \phi_{j,\varepsilon}$ of width $\varepsilon \rho$ centred at the positions $z_j = \mu(t_j)$ that the midpoint mean reaches at the target times t_j , under the conserved installed budget $\int \kappa_w dc = B$ [Eq. (3)]. The deterministic relative trajectory has a strict margin inside the contact tube; the stochastic R_t may leave and re-enter it, with transverse directions on the torus $\mathbb{T}_W^{d-1}$ (right). (b) In the theorem regime $0 < B < B_{\text{top}}^{\text{unif}}(\varepsilon)$, each slab acts as a separated exposure clock: the reaction-time density develops one peak near each passage time t_j (dashed guides), and one valley between adjacent peaks, so $f_{B,w}/B$ has exactly m maxima and $m - 1$ minima on $[\tau, T]$ [Theorem 1]. (Diagram code revised with Codex assistance; scientific content author-verified.)

The budget is measured along the encounter-centre coordinate alone, not over the complete two-particle configuration space: the contact profile χ_a is fixed and common to every design, so Eq. (3) fixes the installed catalyst amount and is the constraint that every admissible allocation w satisfies. In the construction of Sec. III there is one slab per prescribed mode. The killed evolution, reaction-time density, and survival probability are

$$\begin{aligned} \partial_t q_{B,w} &= (\mathcal{L} - K_{B,w})q_{B,w}, & q_{B,w}(0) &= q_0, \\ f_{B,w}(t) &= \langle K_{B,w}, q_{B,w}(t) \rangle, & -S'_{B,w}(t) &= f_{B,w}(t). \end{aligned} \quad (4)$$

Equation (4) is the usual mass balance for a Doi volume-reaction process [15].

For longitudinal catalyst slabs, translation invariance of the common transverse coordinate gives an exact quotient

$$\mathcal{Q}_d^\infty = \mathbb{R}_z \times \mathbb{R}_{r_\parallel} \times \mathbb{T}_W^{d-1}. \quad (5)$$

The Supplemental Material also uses the bounded counterpart $\mathcal{Q}_d^{\text{box}}$. In the construction below (Fig. 1) the midpoint Z_t moves monotonically in mean, the relative

process remains a positive distance inside the contact region along its deterministic trajectory on the chosen window, and each normalized slab is centred at the midpoint mean at a target time. The $W^{-(d-1)}$ factor in the killing field accounts for the omitted common transverse directions and preserves Eq. (3). The Ornstein–Uhlenbeck transport with stationary initial variance is chosen so that $\text{Var } Z_t = \varepsilon^2 D_0 / (2\gamma)$ is constant on the window: the exposure clock is then a common-variance Gaussian mixture in the monotone coordinate $x(t)$, for which the derivative-zero count of Sec. III is exhaustive. The single parameter ε sets both the noise, $D = \varepsilon^2 D_0$, and the slab width $\varepsilon \rho$, so the mixture width $\sigma \propto \varepsilon$ shrinks while the mean trajectory is unchanged.

III. PRESCRIBED FINITE MODALITY: MODEL AND THEOREM

The conserved-budget support construction admits encounter processes with any prescribed *fixed finite* number of reaction-time modes. We state this existence result in an exact quotient of a two-particle Doi process; the construction depends on the prescribed dimension and mode count and does not assert mode-count switching by weight redistribution on one fixed support.

Fix finite $d \geq 2$, finite $m \geq 1$, a transverse period $W > 0$, a contact radius $0 < a < W/2$, positive scales $\ell_0, \rho > 0$, and a compact interval $I = [\tau, T] \subset (0, \infty)$. On $\mathbb{R} \times \mathbb{R} \times \mathbb{T}_W^{d-1}$, consider the independent midpoint and relative processes

$$\begin{aligned} dZ_t &= -\gamma(Z_t - \bar{z})dt + \varepsilon\sqrt{D_0}dW_t, \\ dR_{\parallel,t} &= -\gamma R_{\parallel,t}dt + 2\varepsilon\sqrt{D_0}dW_{\parallel,t}, \\ dR_{\perp,t} &= 2\varepsilon\sqrt{D_0}dW_{\perp,t} \pmod{W}. \end{aligned} \quad (6)$$

The mutually independent initial laws satisfy

$$\begin{aligned} Z_0 &\sim N(z_0, \varepsilon^2 D_0 / (2\gamma)), & D_0, \gamma, u_0 &> 0, & z_0 &\neq \bar{z}, \\ R_{\parallel,0} &\sim N(r_{\parallel,0}, \varepsilon^2 u_0^2), & u_0^2 &< 4D_0/\gamma, & \Sigma_{\perp,0} &\succ 0. \\ R_{\perp,0} &\sim \text{WN}(r_{\perp,0}, \varepsilon^2 \Sigma_{\perp,0}), \end{aligned} \quad (7)$$

Here WN is the wrapped Gaussian on $\mathbb{T}_W^{d-1}$, $\Sigma_{\perp,0}$ is a positive-definite $(d-1) \times (d-1)$ covariance matrix, and $|r|_{\text{mi}}$ is the Euclidean norm of the longitudinal relative coordinate together with the shortest transverse representatives modulo W .

All driving Brownian motions are also independent. Thus $\mu(t) = \bar{z} + (z_0 - \bar{z})e^{-\gamma t}$ is strictly monotone and $\text{Var } Z_t = \varepsilon^2 D_0 / (2\gamma)$: the midpoint is prepared with its stationary variance so that only its mean relaxes, which makes every exposure clock below have the same width σ . The inequality $u_0^2 < 4D_0/\gamma$ is used only to place the initial law in the weighted state space of the transfer estimate in the Supplemental Material. Under the minimum-image convention, assume the deterministic relative trajectory

$r_*(t) = (r_{\parallel,0}e^{-\gamma t}, r_{\perp,0})$ obeys $\sup_{t \in I} |r_*(t)|_{\text{mi}} \leq a - \eta_*$ for some $\eta_* > 0$.

Choose target times

$$\begin{aligned} \tau < t_1 < \dots < t_m < T, \quad c_j = x(t_j), \\ x(t) = \frac{\text{sgn}(\mu')\mu(t)}{\ell_0}, \end{aligned} \quad (8)$$

so that $x'(t)$ is bounded away from zero. At the corresponding physical centres, place normalized Gaussian catalyst slabs of width $\varepsilon\rho$. Their nonnegative weights belong to a compact subset of the simplex,

$$\mathcal{W}_{w_*} = \left\{ \mathbf{w} : \sum_{j=1}^m w_j = 1, \quad 0 < w_* \leq \frac{1}{m} \right\}, \quad (9)$$

and define

$$\begin{aligned} \phi_{j,\varepsilon}(z) &= \frac{\exp[-(z - \mu(t_j))^2 / (2\varepsilon^2 \rho^2)]}{\sqrt{2\pi} \varepsilon \rho}, \\ K_{B,\mathbf{w},\varepsilon}(Z, R) &= \mathbf{1}_{\{|R|_{\text{mi}} < a\}} \frac{B}{W^{d-1}} \sum_{j=1}^m w_j \phi_{j,\varepsilon}(Z). \end{aligned} \quad (10)$$

The slab normalization makes B the same installed centre-space budget for every allocation. Set $K_{B,\mathbf{w},\varepsilon} = BV_{\mathbf{w},\varepsilon}$, and let $T_0(t)$ and q_0 denote the unkilld semigroup and initial law. Then

$$\begin{aligned} G_{\varepsilon,\mathbf{w}}(t) &= B^{-1} \mathbb{E}_0[K_{B,\mathbf{w},\varepsilon}(Z_t, R_t)] = \langle V_{\mathbf{w},\varepsilon}, T_0(t)q_0 \rangle, \\ f_{B,\varepsilon,\mathbf{w}}(t) &= \mathbb{E}_0 \left[K_{B,\mathbf{w},\varepsilon}(Z_t, R_t) e^{-\int_0^t K_{B,\mathbf{w},\varepsilon}(Z_s, R_s) ds} \right]. \end{aligned} \quad (11)$$

When the noise scale is displayed we write $f_{B,\varepsilon,\mathbf{w}}$ for the density $f_{B,\mathbf{w}}$ of Eq. (4). The expectation $\mathbb{E}_0$ is with respect to the unkilld process started from Eq. (7). Independence gives the exact factorization

$$\begin{aligned} G_{\varepsilon,\mathbf{w}}(t) &= a_\varepsilon(t) H_{\sigma,\mathbf{w}}(x(t)), \\ a_\varepsilon(t) &= \frac{c_{d,\varepsilon}(t)}{W^{d-1} \sqrt{2\pi} \varepsilon \sqrt{D_0/(2\gamma) + \rho^2}}, \end{aligned} \quad (12)$$

where $c_{d,\varepsilon}(t) = \mathbb{P}_0(|R_t|_{\text{mi}} < a)$ and $H_{\sigma,\mathbf{w}}$ is defined below. All dependence on d enters through $c_{d,\varepsilon}$ and the normalization $W^{-(d-1)}$.

Theorem 1 (exact prescribed finite modality).— Fix finite $d \geq 2$, finite $m \geq 1$, the data above, and the compact window I . (i) There is $\varepsilon_0 > 0$ such that, for every $0 < \varepsilon < \varepsilon_0$ and every $\mathbf{w} \in \mathcal{W}_{w_*}$, the unit-budget free-exposure clock $G_{\varepsilon,\mathbf{w}}$ has exactly m nondegenerate interior maxima and $m - 1$ nondegenerate interior minima on I , ordered alternately, and no stationary endpoint. (ii) For each such ε there is a budget $B_{\text{top}}^{\text{unif}}(\varepsilon) > 0$ such that the budget-rescaled Doi reaction-time density $f_{B,\varepsilon,\mathbf{w}}/B$ has the same complete stationary signature on I for every $0 < B < B_{\text{top}}^{\text{unif}}(\varepsilon)$ and every $\mathbf{w} \in \mathcal{W}_{w_*}$. Multiplication by B leaves the stationary points unchanged, so the conclusion also holds for the physical density $f_{B,\varepsilon,\mathbf{w}}$ (Theorem S3.8 of the Supplemental Material [49]).

Precisely, for each fixed such ε and allocation set $\mathcal{A} \subseteq \mathcal{W}_{w_*}$, define the topology-retention budget

$$\begin{aligned} B_{\text{top}}(\varepsilon; \mathcal{A}) &:= \sup\{b > 0 : \\ &\quad f_{b,\varepsilon,\mathbf{w}}/b \text{ has the stated} \\ &\quad \text{complete stationary signature on } I \\ &\quad \text{for every } \mathbf{w} \in \mathcal{A} \text{ and every } 0 < \beta < b\}. \end{aligned} \quad (13)$$

Then

$$B_{\text{top}}^{\text{unif}}(\varepsilon) := B_{\text{top}}(\varepsilon; \mathcal{W}_{w_*}) > 0. \quad (14)$$

and (ii) holds with this value.

The mechanism can be seen without invoking a normal form. Gaussian convolution reduces the longitudinal part of the unit-budget exposure to

$$\begin{aligned} H_{\sigma,\mathbf{w}}(x) &= \sum_{j=1}^m w_j \exp\left[-\frac{(x - c_j)^2}{2\sigma^2}\right], \\ \sigma &= \frac{\varepsilon}{\ell_0} \sqrt{\frac{D_0}{2\gamma} + \rho^2}. \end{aligned} \quad (15)$$

Writing $q_j(x) = w_j \exp[-(x - c_j)^2 / (2\sigma^2)]$, $\pi_j(x) = q_j(x) / H_{\sigma,\mathbf{w}}(x)$, and $L = \partial_x \log H$, let $\bar{c}(x)$ and $\text{Var}_{\pi(x)}(c)$ denote the mean and variance of the centres under the finite probability vector $\pi(x)$. Then

$$L(x) = \frac{\bar{c}(x) - x}{\sigma^2}, \quad L'(x) = \frac{\text{Var}_{\pi(x)}(c)}{\sigma^4} - \frac{1}{\sigma^2}. \quad (16)$$

The univariate upper bound of m modes for an m -component Gaussian mixture is classical [41, 42]. Here a direct affine-exponential proof is retained because the Doi transfer needs all critical zeros counted with multiplicity together with uniform root-tube, complement, curvature, and endpoint margins.

For the adjacent pair $c_j < c_{j+1}$, the weighted crossover is

$$s_j = \frac{c_j + c_{j+1}}{2} + \frac{\sigma^2}{c_{j+1} - c_j} \log \frac{w_j}{w_{j+1}}. \quad (17)$$

The adjacent ratio obeys $q_{j+1}/q_j = 1/9$ at $s_j - \sigma^2 \log(9)/(c_{j+1} - c_j)$ and $q_{j+1}/q_j = 9$ at the corresponding plus edge. Thus both adjacent components remain uniformly present throughout the crossover, while all nonadjacent components are exponentially small. Equation (16) then yields one simple minimum at

$$r_j = s_j + O(\sigma^4), \quad L'(r_j) = \frac{(c_{j+1} - c_j)^2}{4\sigma^4} + O(\sigma^{-2}). \quad (18)$$

Each centre similarly contributes one simple maximum at $p_j = c_j + O(\exp[-q/\sigma^2])$, with $L'(p_j) = -\sigma^{-2} + o(\sigma^{-2})$. These constructions give $2m - 1$ simple stationary points. They are exhaustive: after multiplication by a positive Gaussian factor, H' is an exponential polynomial

$$\sum_{j=1}^m (a_j + b_j x) e^{\lambda_j x}, \quad \lambda_1 < \dots < \lambda_m, \quad (19)$$

which has at most $2m - 1$ real zeros counted with multiplicity. The extreme affine-exponential terms dominate in the two tails, so all zeros lie in a compact interval and real analyticity makes their number finite. A generalized Rolle induction then proves the bound by differentiating twice to remove the lowest-exponent affine term.

The remaining relative-coordinate contact probability is a positive slow factor. Uniform bounds on its first two logarithmic time derivatives, peak boxes of width $O(\sigma^2)$, valley boxes of width $O(\sigma^4)$, and a sign argument on the complement of these boxes (the posterior-sector certificate of the Supplemental Material: a sign bound on the logarithmic slope L obtained from the relative weights $\pi_j(x)$ of the slabs) exclude any further stationary point. Consequently the slow factor moves maxima by $O(\sigma^2)$ and minima by $O(\sigma^4)$ without changing their number or type. Finally, on every fixed positive ε , convergence of the budget-rescaled density together with its first two time derivatives, uniformly in the slab weights (Sec. III A),

$$\sup_{\mathbf{w} \in \mathcal{W}_{w_*}} \left\| \frac{f_{B,\varepsilon,\mathbf{w}}}{B} - G_{\varepsilon,\mathbf{w}} \right\|_{C^2(I)} \rightarrow 0 \quad (B \downarrow 0) \quad (20)$$

preserves the disjoint neighbourhoods of the stationary points (root tubes), their signed curvature margins, the derivative margin on their complement, and the endpoint slopes.

The quantifier order is essential: the geometry and admissible scales may depend on fixed (d, m) ; one first fixes a sufficiently small $\varepsilon > 0$ and only then chooses $B < B_{\text{top}}^{\text{unif}}(\varepsilon)$. The theorem does not provide a useful lower bound for that threshold, a uniform-in- d or uniform-in- m construction, an event-mass floor, or a stationary-point count outside I ; it also uses a whole-window contact-interior condition whose small-noise limit is asymptotically saturated contact. Nontrivial contact, observability at a common positive budget, and finite-parameter robustness remain separate numerical checks.

A. Positive-budget transfer

Write $K_{B,\mathbf{w}} = BV_{\mathbf{w}}$ and introduce interior affine control coordinates $\mathbf{w}(\boldsymbol{\theta}) = \tilde{\mathbf{w}} + P\boldsymbol{\theta}$. The budget-rescaled reaction density and the free-exposure law are

$$F_B(t, \boldsymbol{\theta}) = B^{-1} f_{B,\mathbf{w}(\boldsymbol{\theta})}(t), \quad G(t, \boldsymbol{\theta}) = \langle V_{\mathbf{w}(\boldsymbol{\theta})}, e^{t\mathcal{L}} q_0 \rangle. \quad (21)$$

Here G is the reactivity $V_{\mathbf{w}}$ averaged against the unkilld configuration density, and the transfer rests on one observation: $f_{B,\mathbf{w}} = BG + O(B^2)$, with a remainder that stays $O(B^2)$ together with finitely many time and weight derivatives on any compact window $[\tau, T] \subset (0, \infty)$, uniformly over the weights [Eq. (22) below]. Peaks and valleys of G are therefore inherited by $f_{B,\mathbf{w}}$ for all sufficiently small B . On either a bounded reflecting/periodic quotient or the reversible natural-decay realization of Eq. (5), assume q_0 belongs to the corresponding state space and $V_{\mathbf{w}}$ is bounded on a complex control neighbourhood of

a compact interior control set. A Dyson expansion followed by time and control Cauchy estimates gives, for every fixed time order r , finite control multi-index α , and $0 \leq B \leq B_{\text{max}}$,

$$\sup_{(t, \boldsymbol{\theta}) \in [\tau, T] \times \Theta} |\partial_t^r \partial_{\boldsymbol{\theta}}^\alpha (F_B - G)| \leq C_{r,\alpha,\tau,T,\Theta} B, \quad (22)$$

with a constant that depends on B_{max} , the initial law, and the profiles but not on B . The Supplemental Material accompanying this article [49] gives the explicit constant and proof. In particular, C^2 convergence on a compact positive-time window preserves disjoint stationary-point neighbourhoods, curvature signs, endpoint slopes, and a nonzero derivative margin on their complement. The estimate is local in positive time and provides neither a useful numerical value of B nor control on the scale $t = O(B^{-1})$.

B. Three budget thresholds

Three thresholds are kept distinct. The theorem concerns the maximal uniform topology-preservation threshold $B_{\text{top}}^{\text{unif}}(\varepsilon)$. For one fixed allocation, the Supplemental Material [49] derives a closed-form sufficient value $B_{\text{cert}}(\varepsilon; \mathbf{w}, \{U_i\})$, from a term-by-term Dyson-expansion estimate (a bound of the budget expansion in the unweighted pairing, which avoids the weighted norm of the initial law), such that every $0 < B < B_{\text{cert}}$ preserves the free-clock signature. Only this sufficient value, along the fixed anchor family used in its evaluation, has a leading-order form, predicted by the endpoint-slope closed form of the Supplemental Material and confirmed numerically at $\varepsilon = 0.1$ and 0.05 ,

$$\log B_{\text{cert}} = -\frac{c_B}{\varepsilon^2} + O\left(\log \frac{1}{\varepsilon}\right), \quad c_B > 0. \quad (23)$$

Here $\{U_i\}$ is the selected system of disjoint neighbourhoods of the stationary points (root tubes); this dependence is suppressed below. The construction proves only $B_{\text{cert}} \leq B_{\text{top}}(\varepsilon; \{\mathbf{w}\})$, not a bound for the threshold uniform over the full allocation family. High-precision, non-interval evaluations using the same relative-coordinate initial law as the simulations give nominal values $\hat{B}_{\text{cert}} \simeq 8.9 \times 10^{-15}$ for the $m = 2$, $\varepsilon = 0.1$ anchor and 2.6×10^{-7} for the corresponding $m = 3$ geometry. By contrast, B_{op} below is a finite-sample crossing of a stated classifier: $B_{\text{op}} \simeq 7.6$ and 3.6 , respectively. It is not identified with either B_{top} or B_{cert} . The gap measures the conservatism of the certificate rather than the reach of the mechanism: B_{cert} is a sufficient bound obtained by controlling every term of a Dyson expansion uniformly over the whole window, whereas the fixed classifier of Sec. IV records the designed mode count at budgets of order one. The gap is a window-endpoint effect rather than a property of the transfer estimate. The certificate must preserve the sign of G' at the left endpoint τ , where the first clock is

$D_L = \mu(\tau) - \mu(t_1)$ away and $|G'(\tau)|$ carries the Gaussian factor $\exp[-D_L^2/(2\ell_0^2\sigma^2)]$ with σ from Eq. (15); hence $c_B = D_L^2/[2(D_0/(2\gamma) + \rho^2)] = 0.304$ at the $m = 2$ anchor ($D_L = 0.955$) and 0.132 for the $m = 3$ geometry. At $\varepsilon = 0.1$ this single factor accounts for thirteen of the fifteen decades and for about six of the seven, respectively. Depletion does not see the endpoint slope, so the conservatism is that of a whole-window sign certificate; a localized or degree-theoretic sign argument, not a sharper transfer estimate, could reduce it. The weighted-norm route of the Supplemental Material is worse by a further 693 decades and is not used. Closing this gap analytically is open.

IV. NUMERICAL REALIZATION

The theorem is an existence statement with a deliberately sequential quantifier order, and by itself does not identify a practically resolved parameter set. This section reports an off-lattice time-discrete simulation of the same continuum model at moderate noise and order-one budgets, designed to test the theorem’s falsifiable finite-parameter implications, including its predicted failure directions. Table I summarizes the construction as the checkable procedure that the campaigns below execute.

A. Method and sanity checks

No spatial lattice is introduced. The midpoint, longitudinal, and transverse coordinates of Eq. (6) are integrated by Euler–Maruyama steps of size $\Delta t = 10^{-3}$, and Doi killing is realized by per-step thinning with probability $1 - \exp(-K_{B,w,\varepsilon}\Delta t)$ evaluated at the current position. All runs use deterministically seeded counter-based random streams, one substream per walker chunk. Two automated checks accompany every configuration: the per-step kill probability must remain in the unit interval (largest value 5.6×10^{-3} over the seventeen production configurations and their Δt -halving twin, and 3.1×10^{-2} over every campaign run, attained at the $m = 2$, $\varepsilon = 0.05$, $B = 8$ threshold probe), and event counts must balance mass exactly (kills plus survivors equal walkers). Separately, a campaign-level zero-budget twin run accumulates about 10^7 contact steps and produces no kills. A Δt -halving twin of the $(m, \varepsilon, B) = (2, 0.1, 1)$ configuration reproduces both peak times at bin resolution and both peak prominences to within 0.3%, and the $d = 3$ run passes a per-bin Δt -halving comparison with maximum standardized bin difference $|z_{\text{bin}}| = 1.67$ (difference of the two histograms’ bin fractions divided by its pooled binomial standard error).

Modality is decided by a fixed operational classifier: reaction times are histogrammed with bin width 0.02 and smoothed with a Gaussian kernel of bandwidth 0.04, and a local maximum counts as a mode only if its topographic prominence (peak minus contour base) exceeds both five

TABLE I. Finite-window design recipe implied by Theorem 1. Each step cites the statement that licenses it; the recipe prescribes modality only inside the declared window I .

Given finite $d \geq 2$ and $m \geq 1$, a window $I = [\tau, T] \subset (0, \infty)$, and target times $\tau < t_1 < \dots < t_m < T$:

(1) Transport. Choose $(\gamma, z_0, \bar{z})$ with $z_0 \neq \bar{z}$ so that the midpoint mean $\mu(t) = \bar{z} + (z_0 - \bar{z})e^{-\gamma t}$ traverses the design range monotonically with $x'(t)$ bounded away from zero on I [Eq. (8)]. On a compact window $x' > 0$ holds automatically; the quantitative point is to keep $T \lesssim$ a few γ^{-1} so $x'(t_m)$ is not exponentially small (late peaks widen as $\sigma/x'(t_j)$); at the $m = 3$ anchors these temporal widths are 0.07, 0.15, 0.50 at $\varepsilon = 0.1$ and 0.14, 0.30, 1.01 at $\varepsilon = 0.2$, the last exceeding the remaining window $T - t_3 = 0.7$, so choose $T - t_m$ larger than the last width to keep the final peak inside I .

(2) Support. Place the slab centres at $c_j = x(t_j)$, i.e. physical centres $\mu(t_j)$ [Eq. (8)].

(3) Width. Choose ε (and ρ) so that σ of Eq. (15) obeys $\min_j(c_{j+1} - c_j) \geq C\sigma$ with $C \approx 2.3$. At this separation the crossover analysis [Eqs. (17)–(18)] yields one valley per adjacent gap with nonadjacent components exponentially small; strongly unbalanced weights shift the crossover by $\sigma^2 \log(w_j/w_{j+1})/(c_{j+1} - c_j)$ and eat into the margin. Theorem 1 guarantees only that some $\varepsilon < \varepsilon_0$ works; $C \approx 2.3$ is a practical proxy, slightly above the equal-weight two-Gaussian bimodality threshold 2σ .

(4) Verify the free clock. Evaluate $G_{\varepsilon,w}(t) = a_\varepsilon(t)H_{\sigma,w}(x(t))$ [Eqs. (12)–(15)] on I —a cheap one-dimensional computation; the contact factor $c_{d,\varepsilon}(t)$ follows from Gaussian and wrapped-Gaussian marginals—and confirm exactly m nondegenerate maxima and $m - 1$ nondegenerate minima, with clear root separation, curvature signs, and endpoint slopes.

(5) Budget. Theorem 1 guarantees the complete signature for $B < B_{\text{top}}(\varepsilon; \mathcal{W}_{w_*})$, where B_{top} is the topology-retention threshold uniform over the declared allocation family; equivalently, $f_{B,\varepsilon,w}/B$ inherits the complete stationary signature of $G_{\varepsilon,w}$ [Theorem 1; Eqs. (22), (20)]. The theorem asserts that this threshold is positive. For one fixed allocation, the Supplemental Material gives a sufficient expression $B_{\text{cert}}(\varepsilon; \mathbf{w}, \{U_i\}) \leq B_{\text{top}}(\varepsilon; \{\mathbf{w}\})$; its displayed decimal evaluations are high-precision floating-point values, not interval certificates. At practical scales, use the operational classifier crossing $B_{\text{op}}(\varepsilon; m, \mathbf{w}, \mathcal{P})$ only as a separate numerical gate and re-verify f directly at the chosen B (Sec. IV H).

Caveat (quantifier order). The choices are sequential: the geometry and admissible scales may depend on (d, m) ; one first fixes $\varepsilon < \varepsilon_0$ and only then $B < B_{\text{top}}(\varepsilon; \mathcal{W}_{w_*})$. Nothing is claimed about modality outside I , about an event-mass floor, or about uniformity of ε_0 or B_{top} in d or m (Secs. IV H and V).

covariance-aware smoothed-Poisson standard deviations (σ_{prom} , Supplemental Material) and 5% of the maximum smoothed height on the window $I = [0.5, 3.5]$. Peak and

contour base are linear statistics of the raw bin counts, so the prominence variance is $\sum_j (A_{p,j} - A_{b,j})^2 C_j / (N\delta)^2$, with A the kernel smoothing weights, C_j the raw counts, N the walker count, and δ the bin width; a peak-only Poisson sigma, which omits the base variance and the peak-base covariance, would only overstate the significance ratio of counted maxima (by factors up to 1.55 across the retained campaign); it was the run-time gate of the simulation driver and is retained in every stored record, but every verdict reported here is the covariance-aware re-evaluation of the stored counts (Supplemental Material).

Post-hoc sensitivity analysis over bandwidths 0.03–0.05 and relative-prominence floors 3%–7% leaves the principal anchors unchanged but changes the verdict in four of the 96 phase-diagram cells adjacent to the reported boundary. Complete method details, configuration tables, retained histogram counts, and additional figures are given in the Supplemental Material [49].

Computational-tool disclosure.—OpenAI Codex Desktop 0.147.0-alpha.6.5 (GPT-5.6-sol, August 2026), with earlier GPT-5-family Codex sessions (July–August 2026), assisted with software and plotting-code drafting, workflow orchestration, analytical derivation and proof checks, adversarial tests, and reproducibility checks. The author set the questions, model, protocols, acceptance criteria, and claims; verified retained outputs using deterministic seeds, machine-readable histogram and classifier records, targeted reruns, and a checksummed archive; and remains responsible for their interpretation and accuracy.

B. Production grid at $m = 2$ and $m = 3$

Seventeen production configurations with 5×10^6 walkers each, plus one Δt -halving twin, realize the slab design of Theorem 1 at $d = 2$. For $m = 2$, all twelve configurations across $\varepsilon \in \{0.1, 0.2\}$, $B \in \{0.5, 1, 2\}$, and weights $(\frac{1}{2}, \frac{1}{2})$ and $(0.7, 0.3)$ are classified as bimodal under the fixed rule, with the weakest counted peak still 72 covariance-aware smoothed-Poisson standard deviations prominent. Event mass is order one throughout: between 0.29 and 0.92 of all walkers react inside the window. The early peak sits within 0.01–0.03 of its target time (histogram bin width 0.02) in every configuration; the late peak is pulled earlier by progressive depletion as the budget grows, by 0.05–0.11 at $B = 0.5$, 0.09–0.19 at $B = 1$, and 0.15–0.29 at $B = 2$.

For $m = 3$ at $\varepsilon = 0.1$, $B = 1$, the fixed classifier identifies three significant maxima for both the uniform and the $(0.5, 0.3, 0.2)$ allocations, with the first two maxima within 0.01–0.03 of the targets $(0.8, 1.6)$, the third pulled earlier by 0.13–0.17 from 2.8 by the same depletion mechanism, and window event mass 0.66–0.74. At $\varepsilon = 0.2$, $B = 1$, both allocations lose the third mode — consistent with the heuristic spacing margin being nearly exhausted: the minimum adjacent centre spacing equals 2.30σ at $\varepsilon = 0.2$, at the $C \approx 2.3$ spacing boundary of Table I, and the exact

free clock’s third-peak prominence there is only 5.1% of the window maximum (4.98% after the classifier’s smoothing), at the classifier’s 5% floor. Reducing the budget to $B = 0.25$ at uniform weights partially restores the third maximum, which is visible at $t = 2.49$ but uncertified: its prominence is $4.8\sigma_{\text{prom}}$, below the five-sigma gate, and 2.7% of the maximum height, below the 5% floor, so the strict classifier records two modes; consistently, the threshold study below certifies no third-mode crossing anywhere in the scanned budget range for this column.

C. Phase diagram and mode-retention boundary

A 96-cell campaign at equal slab weights scans $\varepsilon \in \{0.05, 0.075, 0.1, 0.125, 0.15, 0.175, 0.2, 0.25\}$ against six budgets $B \in \{0.125, 0.25, 0.5, 1, 2, 4\}$ for both designs under the stated operational rule (Fig. 2). The $m = 2$ design is classified as bimodal in all 48 cells (Fig. S1 of the Supplemental Material). The $m = 3$ design retains all three classifier-significant modes at every budget for $\varepsilon = 0.05$; loses only the largest budget $B = 4$ for $\varepsilon = 0.075$ –0.125; loses $B \geq 2$ at $\varepsilon = 0.15$ and $B \geq 1$ at $\varepsilon = 0.175$; and fails at every budget for $\varepsilon = 0.2$ (the classifier-floor column) and $\varepsilon = 0.25$, where the semi-analytic free clock itself retains only two window maxima. In the first failing cell of each column for $\varepsilon \leq 0.175$ the excluded late maximum is still statistically detectable in the refined 3×10^6 -walker runs—from $6.5\sigma_{\text{prom}}$ at $(\varepsilon, B) = (0.175, 1)$ to $237\sigma_{\text{prom}}$ at $(0.075, 4)$ —but lies at 1.7–4.7% of the window maximum, below the 5% relative-prominence floor; along the displayed boundary the deterministic floor, not the five-sigma gate, is therefore the binding criterion, whereas in the $\varepsilon = 0.2$ column the late candidate fails both conditions at 3×10^6 walkers. The observed staircase is monotone in both variables, and the B -independent loss of the third free-clock maximum is bracketed to $\varepsilon \in (0.2, 0.25]$, overlaid on the figure.

D. Operational budget thresholds

For each ε , an equal-weight bisection chain measures the protocol-defined crossing $B_{\text{op}}(\varepsilon; m, \mathbf{w}_{\text{eq}}, \mathcal{P})$ at which the last mode changes its classifier verdict (Fig. 3). The two designs behave qualitatively differently. For $m = 2$ the measured thresholds are $B_{\text{op}} = 6.99$ at $\varepsilon = 0.05$ and $B_{\text{op}} = 7.62$ at $\varepsilon = 0.1$, and the threshold exceeds the scan limit 8 at $\varepsilon = 0.15$ and 0.2: a mildly increasing trend. For $m = 3$ the operational crossings fall steeply, 4.15, 3.57, and 1.55 at $\varepsilon = 0.05, 0.1$, and 0.15; at $\varepsilon = 0.2$ no probe in the scanned budget range $[0.125, 8]$ certifies a third mode, so no operational crossing exists there under the declared rule.

One remark is required at $(m, \varepsilon) = (3, 0.2)$. There the exact free clock’s third-peak prominence is 5.1% of the window maximum before smoothing and 4.98% after the classifier’s kernel smoothing, so the classified zero-

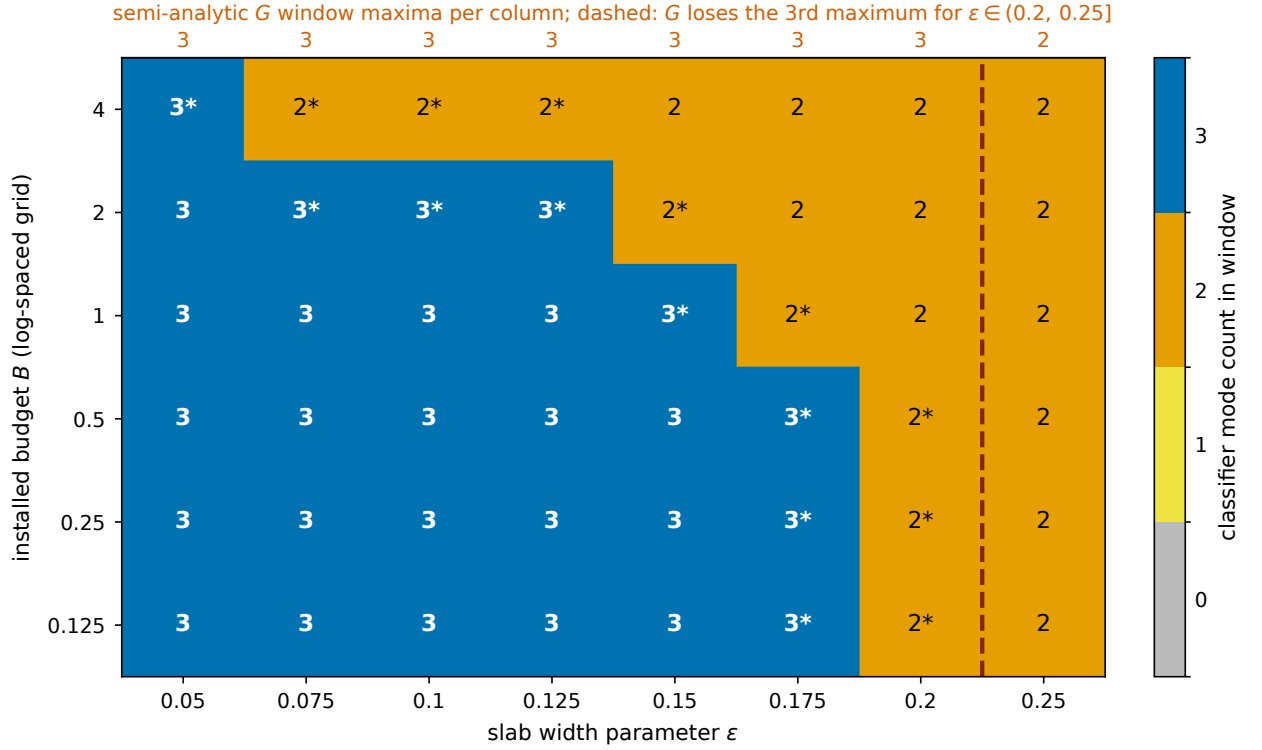


FIG. 2. Operational modality phase diagram for the $m = 3$ design at equal slab weights: colour and printed number give the classifier mode count in each (ε, B) cell (10^6 walkers; * = verdict-boundary-adjacent cells re-run at 3×10^6 walkers, none of the sixteen changing verdict); blue marks cells retaining all three modes, orange cells that lost one. The retained-mode boundary is a monotone staircase in (ε, B) ; the dashed line marks the B -independent semi-analytic boundary $\varepsilon \in (0.2, 0.25]$ beyond which the free exposure clock G itself loses its third window maximum. The $\varepsilon = 0.2$ column sits at the classifier floor (Sec. IVD). (Plotting code revised with Codex assistance; numerical content author-verified.)

budget limit itself sits at the 5% floor: the absence of a certified crossing in this column is a property of the $\varepsilon = 0.2$ design under the declared rule, not of the walker count, and the column probes the classifier floor itself. Under the covariance-aware statistic the two independent streams agree: the phase-diagram grid records two classified modes at every budget in this column, and every threshold probe fails the five-sigma rule (the closest candidates, at $B \leq 0.28$, reach only $4.2\text{--}4.4\sigma_{\text{prom}}$ at relative prominences of 5.3–7.5%). Larger samples do not rescue the column: in the independent phase-grid runs at $B = 0.125$ the late candidate has $3.3\sigma_{\text{prom}}$ at 5.9% relative prominence with 10^6 walkers and $3.4\sigma_{\text{prom}}$ at 3.5% with 3×10^6 , and the 3×10^6 -walker cells at $B = 0.25$ and 0.5 give $2.6\sigma_{\text{prom}}$ at 1.9% and $2.0\sigma_{\text{prom}}$ at 0.9%; at the sample sizes scanned the candidate’s relative prominence falls below the 5% floor as the estimate sharpens, so the absent certification is a classifier-floor outcome rather than a sampling-power limitation. A peak-only sigma convention, which omits the base variance and the peak-base covariance, would instead pass three probes in this column at a nominal $5.8\text{--}6.1\sigma_{\text{prom}}$ and would manufacture both an apparent crossing near $B = 0.28$ and an unexplained disagreement between the two streams; the covariance-aware statistic removes both. The remaining

seven chains are unaffected: every bracket-defining probe verdict is unchanged under the corrected statistic. Their numerical crossings remain classifier-dependent. Across the nine post-hoc classifier settings, bracketed midpoint spans include 6.17–7.83 for $(m, \varepsilon) = (2, 0.1)$ and 3.08–3.63 for $(m, \varepsilon) = (3, 0.1)$, with some settings censored by the stored probe ranges; complete spans and censoring counts are reported in the Supplemental Material.

a. *Mean-field prediction of the retention boundary.* The free clock also predicts where the modes are lost. Replacing the random exposure of a walker by its mean and neglecting the correlation between a walker’s past exposure and its present position gives the hazard–survival law

$$f_1(t) = B G(t) e^{-B\Lambda(t)}, \quad \Lambda(t) = \int_0^t G(s) ds, \quad (24)$$

which has no adjustable parameter. Applying the fixed classifier and pass rule to f_1 (expected counts at 10^6 walkers, no sampling noise) gives $B_{\text{op}}^{\text{mf}} = 7.36, 7.48, > 8, > 8$ for $m = 2$ and 4.29, 3.63, 1.53, none for $m = 3$ at $\varepsilon = 0.05, 0.1, 0.15, 0.2$, against the measured crossings above; the largest deviation, 5% at $(2, 0.05)$, lies inside the post-hoc classifier spans reported in the Supplemental Material. The same evaluation reproduces the classifier verdict in 95

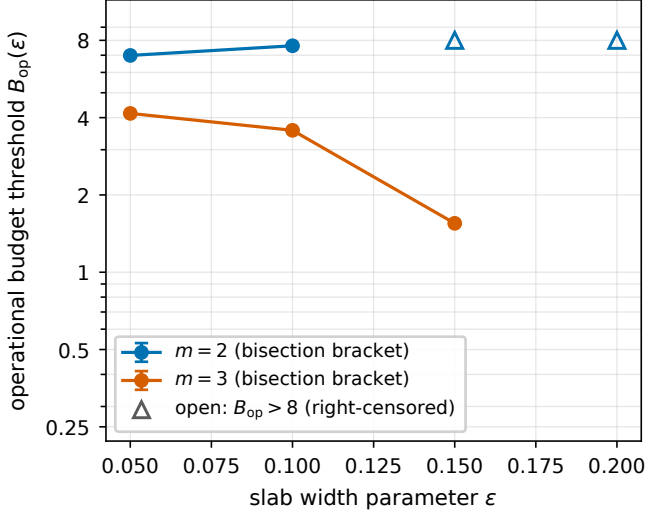


FIG. 3. Empirically bisected operational budget crossing $B_{op}(\epsilon; m, w_{eq}, \mathcal{P})$ for equal slab weights and the fixed protocol $\mathcal{P}$. The crossing is determined by the stated last-mode pass rule at 10^6 walkers per probe and is protocol-dependent; the geometric bisection brackets (Table S2 of the Supplemental Material), of half-width at most 0.04, are narrower than the marker size and are not drawn. They are the deterministic bisection resolution (relative half-width below 0.6%), not sampling intervals; for $(m, \epsilon) = (3, 0.1)$ three independent 10^6 -walker seeds give 3/3/3 modes at $B = 3.50$ and 2/2/2 at $B = 3.64$, confining the seed-level crossing to $(3.50, 3.64)$, within 0.07 of the reported value. For $m = 2$ the threshold is mildly increasing in ϵ and right-censored at the scan limit $B = 8$ beyond $\epsilon = 0.1$; for $m = 3$ it falls steeply as the design margin shrinks, and at $\epsilon = 0.2$ no scanned budget certifies the third mode, so no crossing is plotted there (see text). (Plotting code revised with Codex assistance; numerical content author-verified.)

of the 96 phase-diagram cells [the exception, $(m, \epsilon, B) = (3, 0.075, 4)$, is one of the four classifier-sensitive cells], reproduces every counted peak time of the production grid to within one 0.02 bin, and passes no budget in the $(3, 0.2)$ column. The two loss mechanisms are explicit in Eq. (24) and account for the opposite ϵ trends. For $m = 2$ the late mode is lost through the relative-prominence floor by depletion, $e^{-B\Lambda}$; the exposure accumulated before the second clock decreases with ϵ because the contact factor decays [$c(\tau)$ falls from 1.00 at $\epsilon = 0.05$ to 0.71 at $\epsilon = 0.2$], so a larger budget is needed to suppress it. For $m = 3$ the third mode is lost by merger: its free-clock prominence falls to 5% of the window maximum as the adjacent spacing shrinks to 2.3σ at $\epsilon = 0.2$, so a smaller depletion tilt suffices. Equation (24) is a preflight estimate, not a theorem: the residual, the exposure fluctuation and exposure-position correlation of the exact Doi process, shifts the operational crossings by at most about 5% at these parameters.

E. Robustness and failure modes

Beyond the theorem's hypotheses, a jitter campaign perturbs the slab centres by independent Gaussian displacements of scale ηs (with s the minimum adjacent centre spacing) and the weights by a Dirichlet($50 \bar{w}$) draw, with 50 replicas of 2×10^5 walkers per point at the $\epsilon = 0.1$, $B = 1$ anchors (Fig. S2 of the Supplemental Material). The target classifier mode count survives in at least 96% of replicas up to $\eta = 0.15$ for both designs, decaying to 46% ($m = 2$) and 40% ($m = 3$) at $\eta = 0.4$; all 50 weight-jitter replicas retain the target classifier mode count for both designs. The observed pattern is predominantly consistent with geometric merger: replica-level classifier verdicts agree with the semi-analytic verdict of each replica's own perturbed geometry in 86–100% of cases, and the mismatches preclude a purely geometric attribution. Classified failures are associated mainly with mergers of adjacent peaks when jittered spacing collapses, rather than with extra peaks.

F. Higher mode count and dimension

Two further campaigns probe the prescribed- m and prescribed- d directions at fixed design effort. For a five-slab design on a stretched geometry, the fixed classifier identifies five significant maxima near the five target times (to within 0.05), with prominences $215\text{--}454\sigma_{\text{prom}}$ and window event mass 0.53 per walker (Fig. 4). A $d = 3$ spot check — two transverse torus directions, killing prefactor B/W^2 , all other parameters unchanged from the $(m, \epsilon, B) = (2, 0.1, 1)$ anchor — is classified with two significant maxima at $t = 0.99$ and 2.37, with window event mass 0.60 per walker (Fig. S3 of the Supplemental Material). This check is consistent with the quotient construction in the first higher dimension; it is not a dimension-uniform numerical result.

G. Consistent dimensionalization of the numerical anchor

Comparison of the generator and stochastic differential equations gives the physical single-particle diffusivity $D = \epsilon^2 D_0$, not D_0 . With length and time units $1 \mu\text{m}$ and 1 s, the $\epsilon = 0.1$, $D_0 = 1$ anchor therefore corresponds to $D = 0.01 \mu\text{m}^2/\text{s}$. This slow-diffusion anchor corresponds to a confined colloid or a particle in a viscous medium. Retaining the simulated $W = 1$ gives a $1 \mu\text{m}$ transverse period and, for the equal-weight $m = 3$, $B = 1$ case, an on-contact single-slab peak rate of 1.33 s^{-1} . This is a dimensionalization of the simulated point, not evidence that $B = 1$ lies below B_{top} ; the full mapping is given in the Supplemental Material. In these units $B_{\text{phys}} = BL_0^d/t_0$ has the dimensions of a d -dimensional bimolecular rate coefficient ($1 \mu\text{m}^2/\text{s}$ at $B = 1$, $d = 2$). The alternative anchor $t_0 = 0.01 \text{ s}$ tabulated in the Supplemental Material

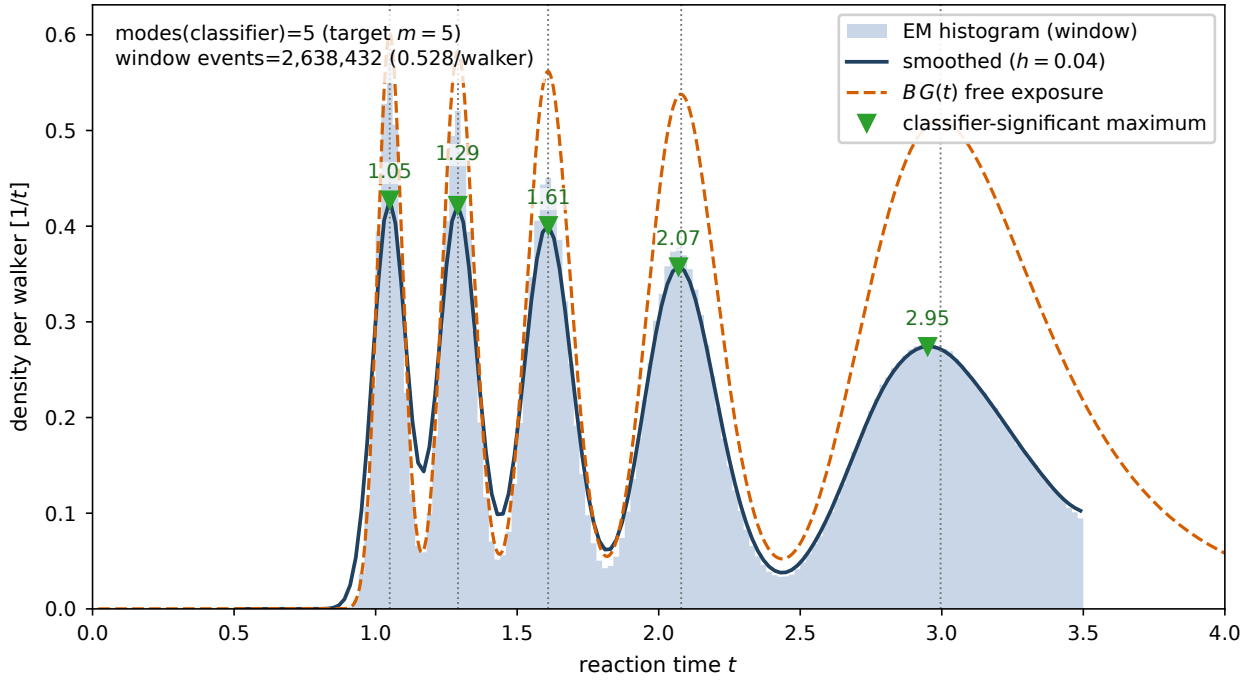


FIG. 4. Five-mode off-lattice realization at $\varepsilon = 0.1$, $B = 1$, uniform weights, on a stretched geometry ($z_0 = 8$; slab centres $\mu_j = (2.8, 2.2, 1.6, 1.0, 0.4)$). Histogram (legend “EM”): Euler–Maruyama simulation with 5×10^6 walkers. The fixed classifier identifies five significant maxima near the design target times (dotted lines), with prominences between 215 and 454 covariance-aware smoothed-Poisson standard deviations (σ_{prom}) and 0.53 reaction events per walker inside the window. Dashed curve: free-exposure prediction $BG(t)$. (Plotting code revised with Codex assistance; numerical content author-verified.)

realizes $D = 1 \mu\text{m}^2/\text{s}$, $\gamma = 100 \text{ s}^{-1}$, a 5–35 ms window, and 100 nm-wide slabs spaced by 0.6–1 μm for the $m = 3$ design at $\varepsilon = 0.1$ (Table S5 of the Supplemental Material); no experimental realization is claimed.

H. Scope of the numerical evidence

The campaigns give finite-parameter evidence that the designed signatures can be observed at moderate noise, order-one budgets, and order-one event mass. They also locate failures in the anticipated directions: depletion at large budget, clock merger as noise or centre jitter grows, and loss of a free-clock maximum when its separation margin closes. These are classifier outcomes, not exact root counts for the time-discretized process.

No spatial lattice is introduced, but the simulations retain both time-step and sampling error. The Δt -halving comparisons are consistency checks, not a convergence theorem. Three independent seeds agree on all 18 verdicts across six targeted configurations, and the $m = 3$ and $m = 5$ interior anchors retain their mode counts when Δt is halved. However, one independent $\Delta t/2$ run of the boundary-adjacent $(m, \varepsilon, B) = (3, 0.175, 0.5)$ cell changes from three to two classified modes because the late candidate peak falls below both classifier thresholds. Since the step size also changes the random-number stream, this comparison does not isolate time-step from sampling er-

ror; the displayed phase-grid verdict is resolution/stream-sensitive, although the six other stored runs of this cell at $\Delta t = 10^{-3}$ (5×10^5 – 3×10^6 walkers, five seeds) all retain three modes (Supplemental Material).

V. DISCUSSION

The theorem connects spatial first-passage theory with inverse timing design: a static, non-switchable catalyst budget of fixed integrated strength suffices to prescribe the complete finite-window peak–valley structure of a reaction-time density, a control that the inverse-design paradigms of Sec. I obtain with time-dependent barriers, gating, or hidden kinetic phases. Relative to strategically arranged absorbing traps, it provides an arbitrary prescribed fixed finite mode count, an exhaustive maxima–minima count on a declared window, uniformity over a compact family of positive allocations, and transfer to a two-particle positive-budget Doi law under a conserved centre-coordinate resource. Relative to inverse first-passage constructions, it uses a static lab-frame field but prescribes only finite-window topology rather than an arbitrary lifetime distribution. Relative to phase-type, gated, and proofreading mechanisms, it introduces no hidden reaction phase or time-dependent switch. The novelty is this conjunction of constraints, not multimodality by itself.

Mathematically, stationary catalyst locations become separated exposure clocks. The affine-exponential zero bound [Eq. (19)] makes the Gaussian-clock count exhaustive; root-tube and posterior-sector estimates retain a derivative margin on the entire complement; the contact factor perturbs rather than creates the clock signature; and compact-positive-time C^2 convergence transfers all maxima, minima, and curvature signs to the killed law. The transfer itself is perturbative in B at fixed ε ; the substantive content is the whole-window certificate with margins uniform over the allocation family and the explicit, weighted-norm-free sufficient budget B_{cert} , neither of which follows from C^2 continuity alone. Uniformity over $w_j \geq w_*$ is important: the conclusion is not tied to one finely tuned allocation, although changing weights on the same support is not claimed to switch among arbitrary mode counts. The budget itself enters the theorem only as the perturbation scale: $K_{B,w} = BV_w$ is linear in B and Theorem 1 is a small- B statement, so the fixed- L^1 constraint defines the comparison class of allocations rather than an obstruction that the proof overcomes; its content lies in the uniformity over that class and, numerically, in the fixed- B comparisons of allocation pairs and weight jitter in Sec. IV.

The simulations provide finite-parameter evidence rather than an additional theorem. At the scanned parameters they recover the designed classifier mode counts, locate operational mode-retention boundaries, and show failure through the same two mechanisms indicated by the proof margins: adjacent-clock merger as noise or geometric jitter grows, and suppression of a late mode by depletion as the budget grows. Both mechanisms, and the operational boundary itself, are reproduced without adjustable parameters by the mean-field hazard-survival law $f_1 = BG e^{-B\Lambda}$ built from the free exposure clock, to within about 5% in the crossings and one bin in the peak times; the residual is the exposure fluctuation and exposure-position correlation of the exact Doi process, which Theorem 1 controls only in the small- B limit. The semi-analytic free clock is consequently a quantitative preflight diagnostic, but the finite-walker classifier, seed, bandwidth, prominence rule, and time step remain part of the operational definition near a boundary.

The restrictions are substantive. First, the deterministic relative trajectory lies strictly inside the contact region and the free contact probability approaches one in the small-noise limit. The modal skeleton is therefore generated primarily by separated longitudinal exposure clocks, not by a nontrivial contact-gating effect. The centre-coordinate dependence of K is itself an idealization: at the $\varepsilon = 0.1$ anchor the slab width $\varepsilon\rho = 0.1$ is smaller than the contact radius $a = 0.4$ and comparable to the stationary longitudinal separation $\varepsilon\sqrt{2D_0/\gamma} = 0.14$, so a static field acting on the position of one particle rather than on the pair centre would convolve each clock with the relative law. Along the deterministic relative trajectory the resulting shift, $|r_*|/2 \leq 0.03$ on I , is well below the slab width, and the added variance $\text{Var } R_{\parallel}/4 \leq \varepsilon^2 D_0/(2\gamma)$

would enlarge σ by about 15% at this anchor; a single-particle field is therefore a natural finite-parameter test beyond the theorem, not part of it. Second, the extension to $d \geq 2$ uses longitudinal slabs and a transverse symmetry quotient; it demonstrates dimension compatibility rather than a new dimension-driven geometry. Third, the normalized slab height scales as $1/\varepsilon$, so the fixed- L^1 constraint imposes neither an independent peak-rate cap nor a minimum feature size; the supremum of the field, $B \max_j w_j / (\sqrt{2\pi} \varepsilon \rho W^{d-1})$ for well-separated slabs, is small in the theorem regime only because B is taken small at fixed ε , and an additional L^∞ cap is binding only at finite parameters (on-contact peak rate 1.33 at the $m = 3, B = 1$ anchor). Fourth, B_{top} is a topology threshold, B_{cert} is only a fixed-allocation sufficient value, and B_{op} is a classifier-dependent finite-sample crossing; none is identified with another.

Nothing is proved about topology outside I , a parameter choice uniform in d or m , a common positive event-mass floor, or mode-count switching on one fixed support family. Natural next questions are whether prescribed modality persists under simultaneous L^1 and L^∞ constraints, under a minimum catalyst width, or when the contact probability remains genuinely nontrivial throughout the design window. Within the stated scope, the result gives a constructive topology theorem for a static resource-constrained Doi field and a finite-parameter test of its mechanism.

ACKNOWLEDGMENTS

The Codex systems and versions identified in Sec. IV A also assisted with literature and cross-document consistency checks and with drafting or revising explanatory text. Retained citations were checked against the cited records, and the manuscript was rebuilt and inspected after revision. The AI tool is not an author; the sole author takes responsibility for the retained content.

DATA AVAILABILITY

The code, parameter files, machine-readable numerical records (including the retained histogram counts, the classifier diagnostics, and the covariance-aware re-judgement of every stored classifier record), figure-generation scripts, and the Lean 4 machine-checked kernel package described in the Supplemental Material are openly available in the versioned reproduction archive of Ref. [50], released as tag v1.0.0 at <https://github.com/zhuyi-xiaoxiao/prescribed-reaction-time-modes>; every file in the archive can be checked against its SHA-256 manifest (MANIFEST.sha256). Raw walker trajectories and unbinned reaction times were not retained; the deterministic simulation scripts and seed metadata needed to regenerate them are included. Source code is released under the

-
- [1] M. von Smoluchowski, Versuch einer mathematischen Theorie der Koagulationskinetik kolloider Lösungen, *Zeitschrift für Physikalische Chemie* **92**, 129 (1917).
- [2] F. C. Collins and G. E. Kimball, Diffusion-controlled reaction rates, *Journal of Colloid Science* **4**, 425 (1949).
- [3] S. Redner, *A Guide to First-Passage Processes* (Cambridge University Press, Cambridge, 2001).
- [4] R. Metzler, G. Oshanin, and S. Redner, eds., *First-Passage Phenomena and Their Applications* (World Scientific, Singapore, 2014).
- [5] Z. Schuss, A. Singer, and D. Holcman, The narrow escape problem for diffusion in cellular microdomains, *Proceedings of the National Academy of Sciences* **104**, 16098 (2007).
- [6] D. Holcman and Z. Schuss, The narrow escape problem, *SIAM Review* **56**, 213 (2014).
- [7] S. Condamin, O. Bénichou, V. Tejedor, R. Voituriez, and J. Klafter, First-passage times in complex scale-invariant media, *Nature* **450**, 77 (2007).
- [8] O. Bénichou and R. Voituriez, From first-passage times of random walks in confinement to geometry-controlled kinetics, *Physics Reports* **539**, 225 (2014).
- [9] P. C. Bressloff and J. M. Newby, Stochastic models of intracellular transport, *Reviews of Modern Physics* **85**, 135 (2013).
- [10] D. S. Grebenkov, R. Metzler, and G. Oshanin, Towards a full quantitative description of single-molecule reaction kinetics in biological cells, *Physical Chemistry Chemical Physics* **20**, 16393 (2018).
- [11] D. S. Grebenkov, Spectral theory of imperfect diffusion-controlled reactions on heterogeneous catalytic surfaces, *The Journal of Chemical Physics* **151**, 104108 (2019).
- [12] T. G. Mattos, C. Mejía-Monasterio, R. Metzler, and G. Oshanin, First passages in bounded domains: When is the mean first passage time meaningful?, *Physical Review E* **86**, 031143 (2012).
- [13] A. Godec and R. Metzler, Universal proximity effect in target search kinetics in the few-encounter limit, *Physical Review X* **6**, 041037 (2016).
- [14] A. Godec and R. Metzler, First passage time distribution in heterogeneity controlled kinetics: going beyond the mean first passage time, *Scientific Reports* **6**, 20349 (2016).
- [15] M. Doi, Stochastic theory of diffusion-controlled reaction, *Journal of Physics A: Mathematical and General* **9**, 1479 (1976).
- [16] E. Teramoto and N. Shigesada, Theory of bimolecular reaction processes in liquids, *Progress of Theoretical Physics* **37**, 29 (1967).
- [17] R. Erban and S. J. Chapman, Stochastic modelling of reaction-diffusion processes: algorithms for bimolecular reactions, *Physical Biology* **6**, 046001 (2009).
- [18] A. J. Bray, S. N. Majumdar, and G. Schehr, Persistence and first-passage properties in nonequilibrium systems, *Advances in Physics* **62**, 225 (2013).
- [19] A. E. Lindsay, R. T. Spoonmore, and J. C. Tzou, Hybrid asymptotic-numerical approach for estimating first-passage-time densities of the two-dimensional narrow capture problem, *Physical Review E* **94**, 042418 (2016).
- [20] A. Godec and R. Metzler, First passage time statistics for two-channel diffusion, *Journal of Physics A: Mathematical and Theoretical* **50**, 084001 (2017).
- [21] D. S. Grebenkov, R. Metzler, and G. Oshanin, Strong defocusing of molecular reaction times results from an interplay of geometry and reaction control, *Communications Chemistry* **1**, 96 (2018).
- [22] D. S. Grebenkov, R. Metzler, and G. Oshanin, Full distribution of first exit times in the narrow escape problem, *New Journal of Physics* **21**, 122001 (2019).
- [23] F. Le Vot, S. B. Yuste, E. Abad, and D. S. Grebenkov, First-encounter time of two diffusing particles in two- and three-dimensional confinement, *Physical Review E* **105**, 044119 (2022).
- [24] C. Zucca and L. Sacerdote, On the inverse first-passage-time problem for a Wiener process, *The Annals of Applied Probability* **19**, 1319 (2009).
- [25] E. Ekström and S. Janson, The inverse first-passage problem and optimal stopping, *The Annals of Applied Probability* **26**, 3154 (2016).
- [26] B. Ettinger, S. N. Evans, and A. Hening, Killed Brownian motion with a prescribed lifetime distribution and models of default, *The Annals of Applied Probability* **24**, 1 (2014).
- [27] B. Ettinger, A. Hening, and T. K. Wong, The inverse first passage time problem for killed Brownian motion, *The Annals of Applied Probability* **30**, 1251 (2020).
- [28] A. Klump and M. Kolb, An elementary approach to the inverse first-passage-time problem for soft-killed Brownian motion, *Journal of Applied Probability* **61**, 279 (2024).
- [29] S. Asmussen, O. Nerman, and M. Olsson, Fitting phase-type distributions via the EM algorithm, *Scandinavian Journal of Statistics* **23**, 419 (1996).
- [30] A. Thümmel, P. Buchholz, and M. Telek, A novel approach for phase-type fitting with the EM algorithm, *IEEE Transactions on Dependable and Secure Computing* **3**, 245 (2006).
- [31] A. Szabo, D. Shoup, S. H. Northrup, and J. A. McCammon, Stochastically gated diffusion-influenced reactions, *The Journal of Chemical Physics* **77**, 4484 (1982).
- [32] P. C. Bressloff and S. D. Lawley, Stochastically gated diffusion-limited reactions for a small target in a bounded domain, *Physical Review E* **92**, 062117 (2015).
- [33] A. Kumar, Y. Scher, S. Reuveni, and M. S. Santhanam, Inference from gated first-passage times, *Physical Review Research* **5**, L032043 (2023).
- [34] J. J. Hopfield, Kinetic proofreading: A new mechanism for reducing errors in biosynthetic processes requiring high specificity, *Proceedings of the National Academy of Sciences* **71**, 4135 (1974).
- [35] J. Ninio, Kinetic amplification of enzyme discrimination, *Biochimie* **57**, 587 (1975).
- [36] T. W. McKeithan, Kinetic proofreading in T-cell receptor signal transduction, *Proceedings of the National Academy of Sciences* **92**, 5042 (1995).
- [37] T. D. Keidar and S. Reuveni, Universal linear response of first-passage kinetics: A framework for prediction and inference, *Physical Review Research* **8**, 023135 (2026).

- [38] B. T. Nguyen and D. S. Grebenkov, A spectral approach to survival probabilities in porous media, *Journal of Statistical Physics* **141**, 532 (2010).
- [39] Z. C. Scott, A. I. Brown, S. S. Mogre, L. M. Westrate, and E. F. Koslover, Diffusive search and trajectories on tubular networks: a propagator approach, *The European Physical Journal E* **44**, 80 (2021).
- [40] K. Nicolaou and B. M. Mulder, A probabilistic algorithm for optimising the steady-state diffusional flux into a partially absorbing body, *Scientific Reports* **13**, 22815 (2023).
- [41] M. A. Carreira-Perpiñán and C. K. I. Williams, On the number of modes of a Gaussian mixture, in *Scale-Space Methods in Computer Vision*, Lecture Notes in Computer Science, Vol. 2695 (Springer, 2003) pp. 625–640.
- [42] C. Améndola, A. Engström, and C. Haase, Maximum number of modes of Gaussian mixtures, *Information and Inference: A Journal of the IMA* **9**, 587 (2020).
- [43] S. A. Isaacson and Y. Zhang, An unstructured mesh convergent reaction–diffusion master equation for reversible reactions, *Journal of Computational Physics* **374**, 954 (2018).
- [44] S. A. Isaacson, A. J. Mauro, and J. Newby, Uniform asymptotic approximation of diffusion to a small target: Generalized reaction models, *Physical Review E* **94**, 042414 (2016).
- [45] T. Prüstel and M. Meier-Schellersheim, Rate coefficients, binding probabilities, and related quantities for area reactivity models, *The Journal of Chemical Physics* **141**, 194115 (2014).
- [46] P. C. Bressloff, Diffusion-mediated absorption by partially-reactive targets: Brownian functionals and generalized propagators, *Journal of Physics A: Mathematical and Theoretical* **55**, 205001 (2022).
- [47] D. S. Grebenkov, Paradigm shift in diffusion-mediated surface phenomena, *Physical Review Letters* **125**, 078102 (2020).
- [48] L. Giuggioli, S. Pérez-Becker, and D. P. Sanders, Encounter times in overlapping domains: Application to epidemic spread in a population of territorial animals, *Physical Review Letters* **110**, 058103 (2013).
- [49] See Supplemental Material at [URL will be inserted by publisher] for the analytic generator realizations, compact-positive-time mixed-jet bound, complete prescribed-modality proof, fixed-allocation sufficient-budget formula, numerical methods, consistency checks, classifier-sensitivity analysis, configuration tables, and supplementary figures of the off-lattice campaigns, which includes Refs. [51, 52].
- [50] X. Zhouyi, Reproduction package for “Prescribing finite-window reaction-time modes with a static fixed-budget Doi reactivity field”, GitHub repository, release v1.0.0, <https://github.com/zhouyi-xiaoxiao/prescribed-reaction-time-modes> (2026), Code: MIT License; data, figures, and documentation: CC BY 4.0.
- [51] D. S. Grebenkov, Joint distribution of multiple boundary local times and related first-passage time problems with multiple targets, *Journal of Statistical Mechanics: Theory and Experiment* **2020**, 103205 (2020).
- [52] X. Zhouyi, Spectral diagnostics and local fixed-budget sensitivity of critical points in finite encounter-reaction models (2026), *J. Chem. Phys.*, submitted (manuscript JCP26-AR-03623).